

Task 1 (Data transformation)

Objective: Prepare and transform data for analysis

1. Removed duplicate Rows based on key column Made sure the data types were correct Handled any missing value
2. Created calculated column Using the following DEX function
 - Membership duration = `DATEDIFF([Membership_Since], TODAY(), DAY)`
 - Transaction year = `YEAR([Transaction_Date])`
 - Transaction month = `MONTH([Transaction_Date])`
3. Created Relationships In view model

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐	From: table (column)	Relationship	To: table (column)	Status	
☐	Churn_Labelled_Customers (1)...		Customer_Demographics (1) (...	Active	...
☐	Customer_Transactions (1) (Cu...		Customer_Demographics (1) (...	Active	...
☐	Customer_Transactions (1) (Cu...		Churn_Labelled_Customers (1)...	Inactive	...
☐	Customer_Transactions (1) (Sto...		Store_Locations (1) (Store_ID)	Active	...
☐	Loyalty_Program (1) (Custome...		Customer_Demographics (1) (...	Active	...

Close

Task 2 (Identifying churn rate)

Objective: Identify and visualize churn rate By various Criteria's

Measures: Below are the measures used in task

measures table

☐ churn_Rate

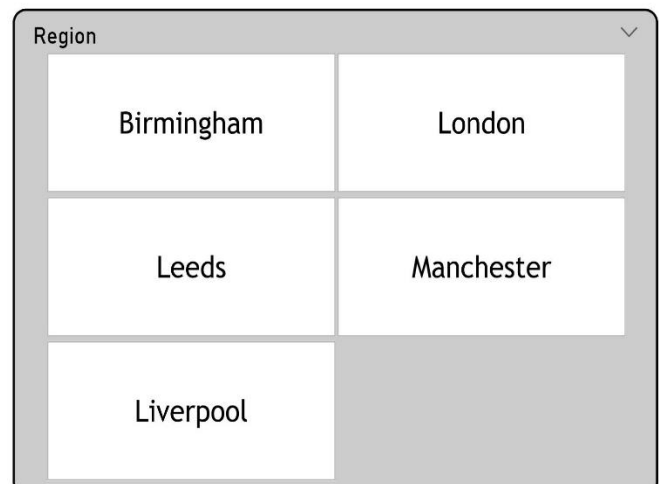
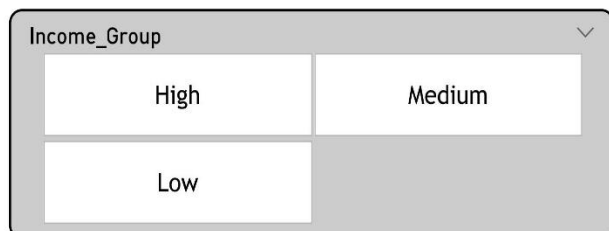
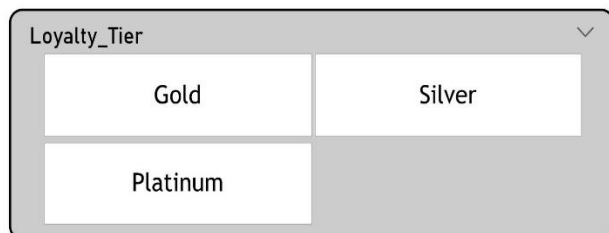
☐ no of Churned labled Customer

☐ no of cutomer in Churn table

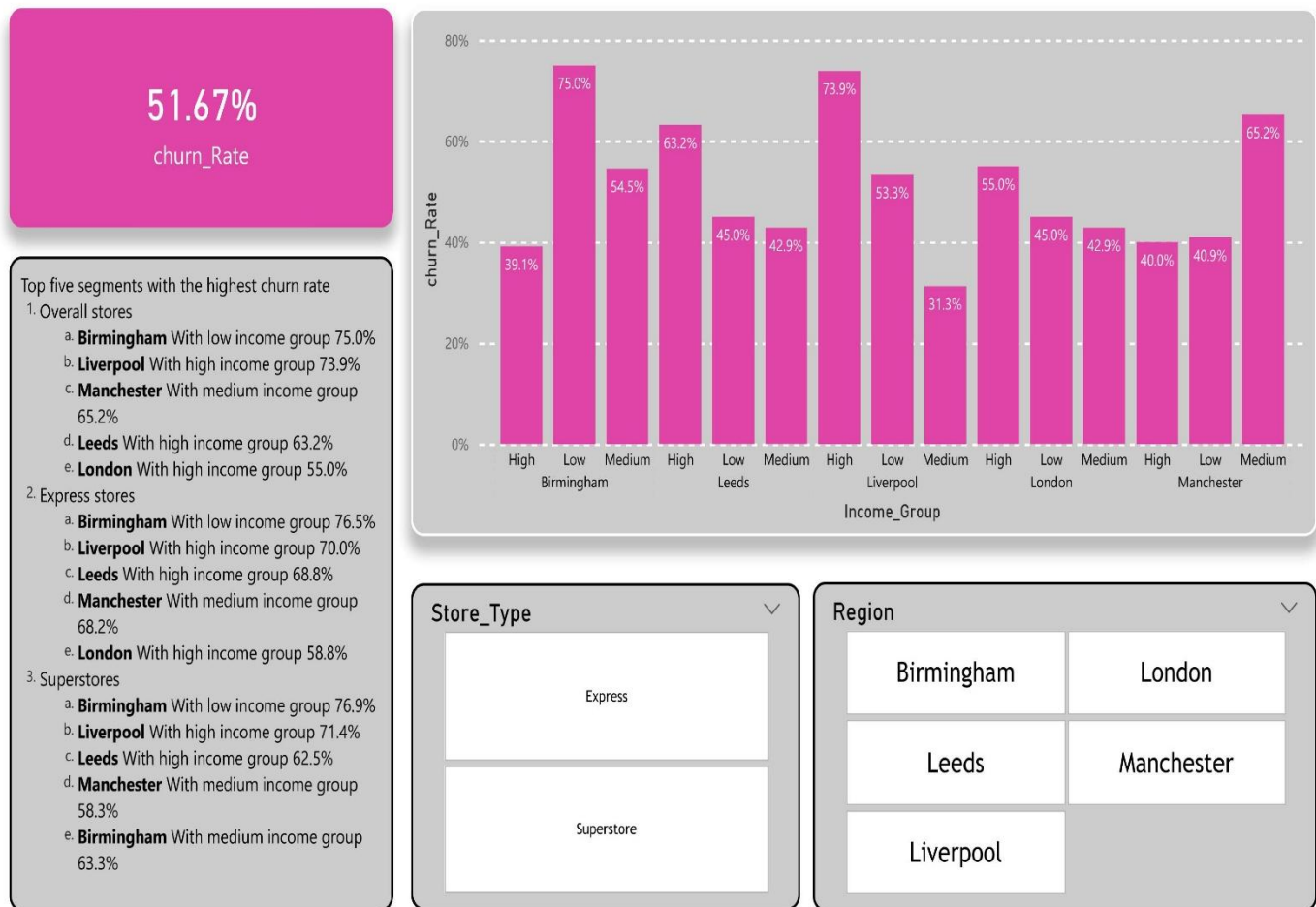
☐ Repeat customer

[Measures](#)

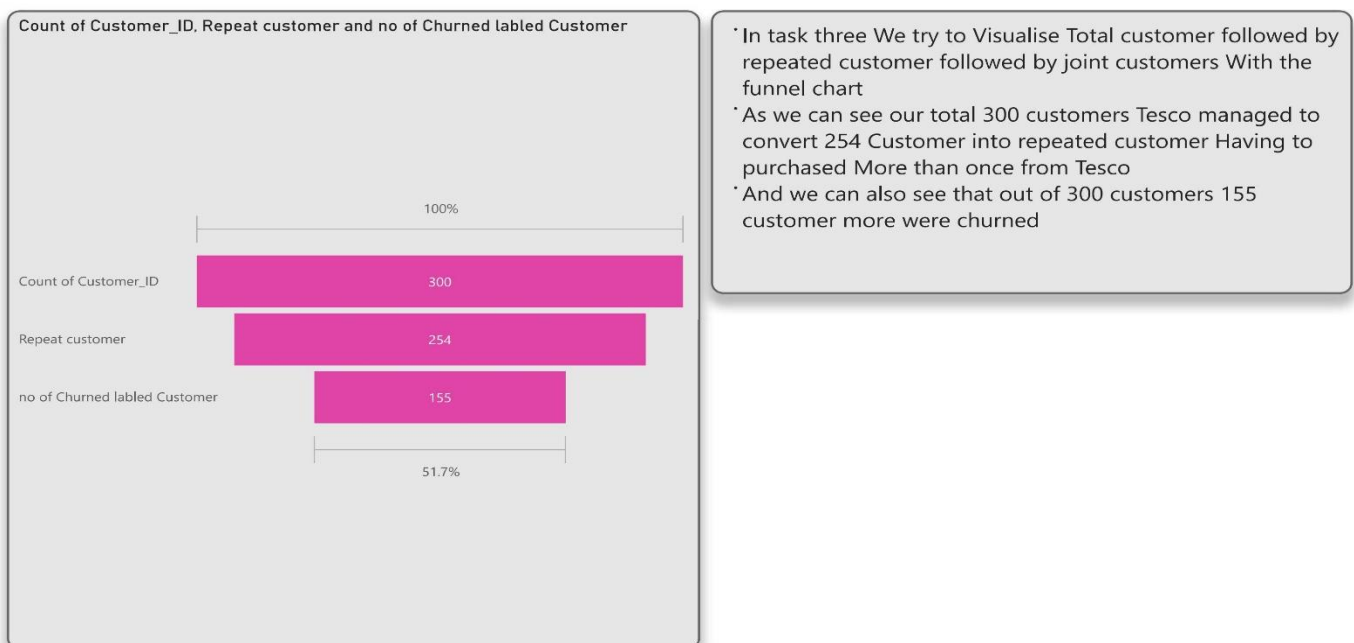
1. Creating churn rate card.



2. Visualizing churn rate.



3. Creating funnel chart.



Task 3 (Customer loyalty and retention)

Objective: Understand customer loyalty True repeated behavior.

Measures: Below are the measures used for This task.

Measure table

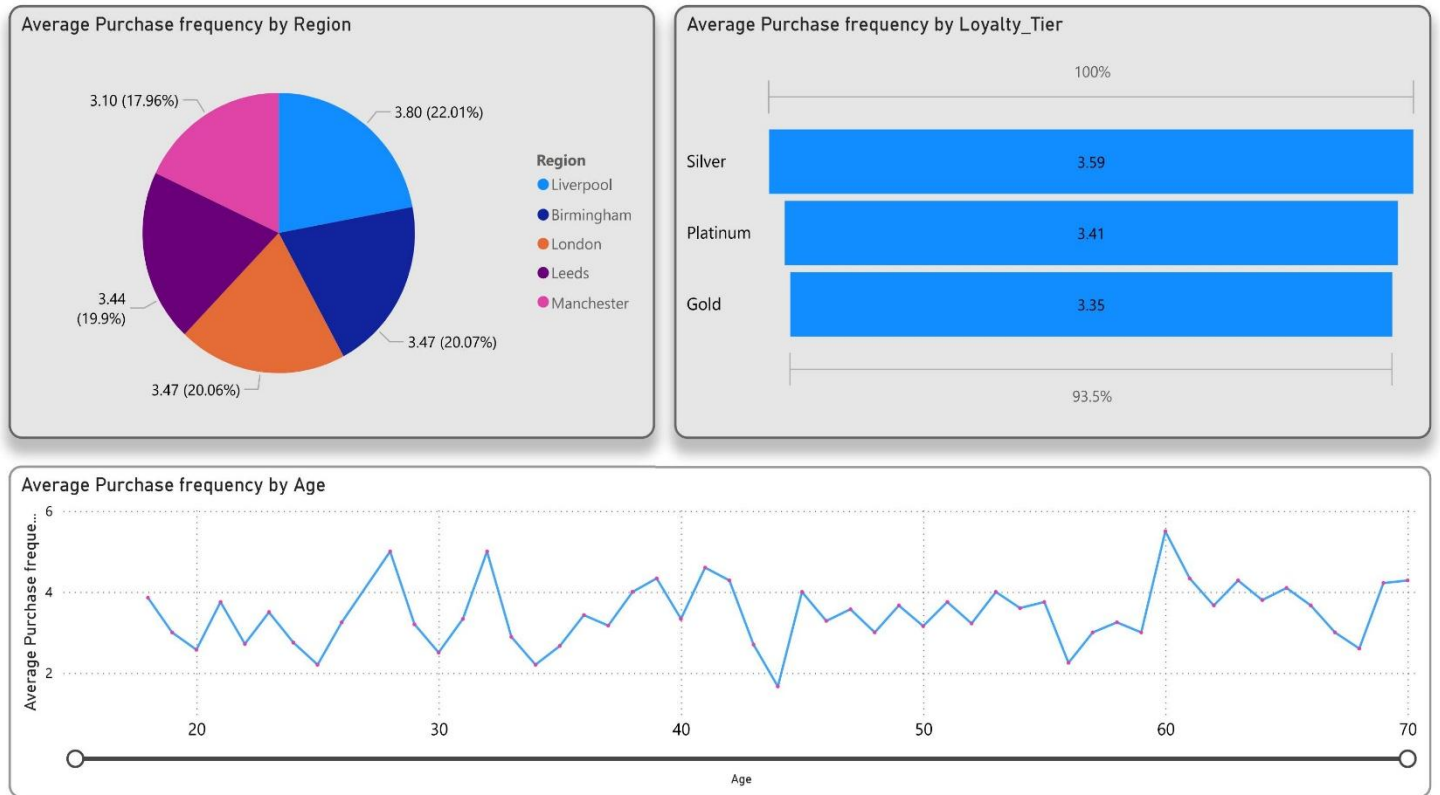
- ☐ Average order quantity
- ☐ Average Purchase frequency
- ☐ High tear customer
- ☐ Low tear customer
- ☐ mid tear customer
- ☐ Repeat customer
- ☐ repeat customer during promo
- ☐ Repeat rate

[Measures](#)

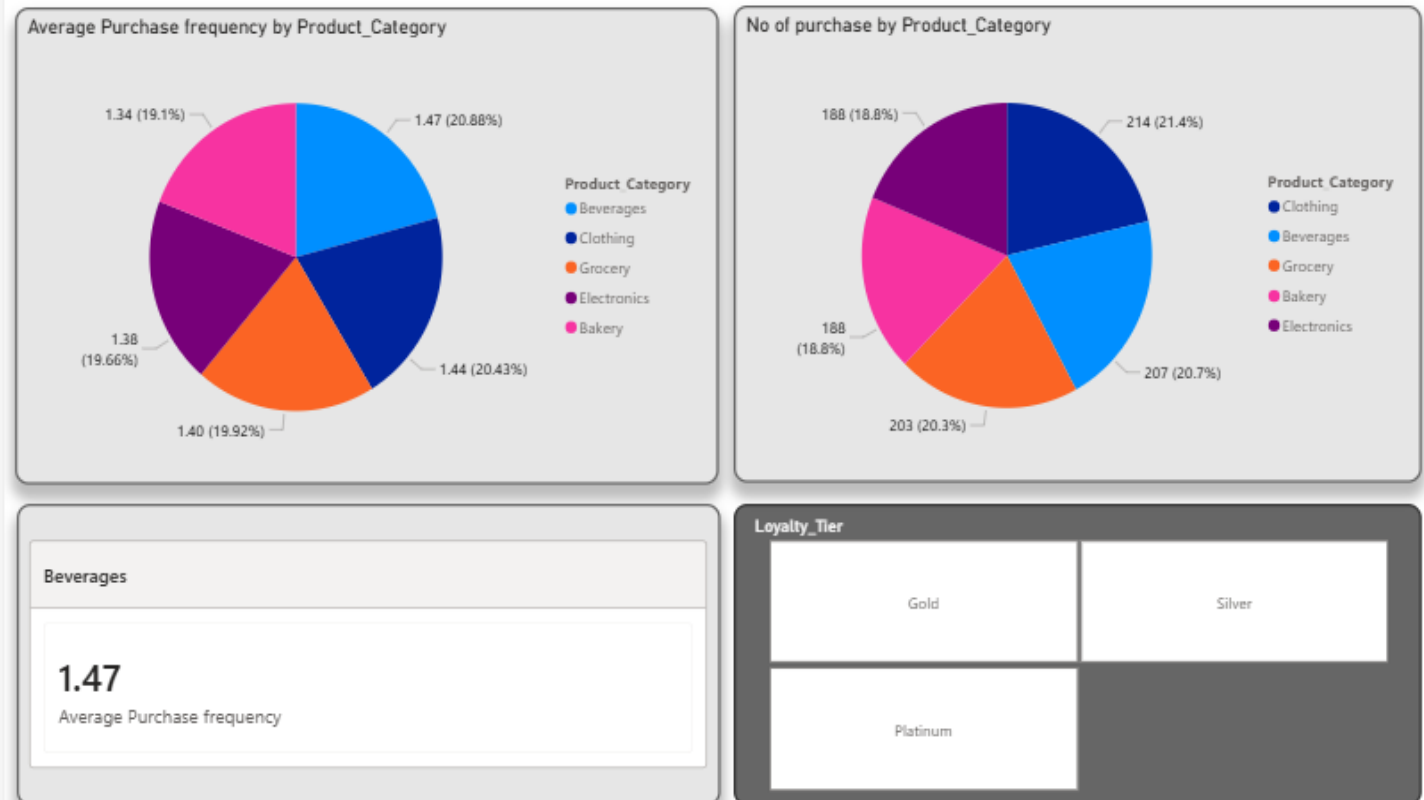
1) Creating measures For low high and medium tier customers.



2) Comparing purchase frequency.



3) Identifying most bought Product category By loyal customers.



Task 4 (Performance and loyalty evaluation)

Objective: Evaluating how Promotion and loyalty tier Affect retention

Measures: Below are the measures used in this task

- Measure table
- ☐ avg purchase amount (no promo)
 - ☐ avg purchase amount (promo)
 - ☐ Churn rate
 - ☐ Churned people
 - ☐ Non Churned people
 - ☐ redemption rate
 - ☐ retention rate
 - ☐ Transaction with promotion
 - ☐ Transaction with promotion %

[Measures](#)

1) Creating measures

51.00%

Transaction with promotion %

248.64

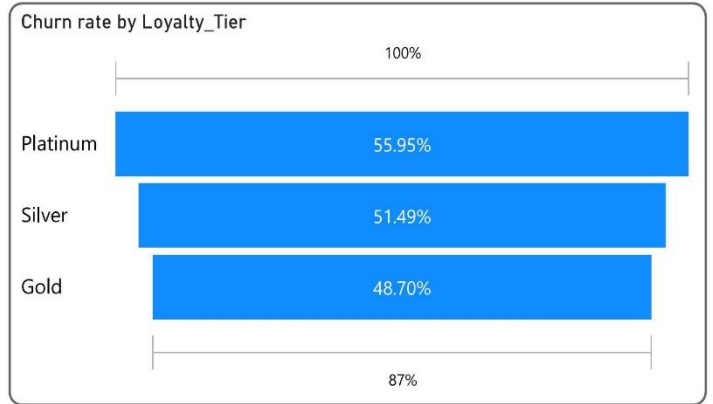
avg purchase amount (no promo)

250.03

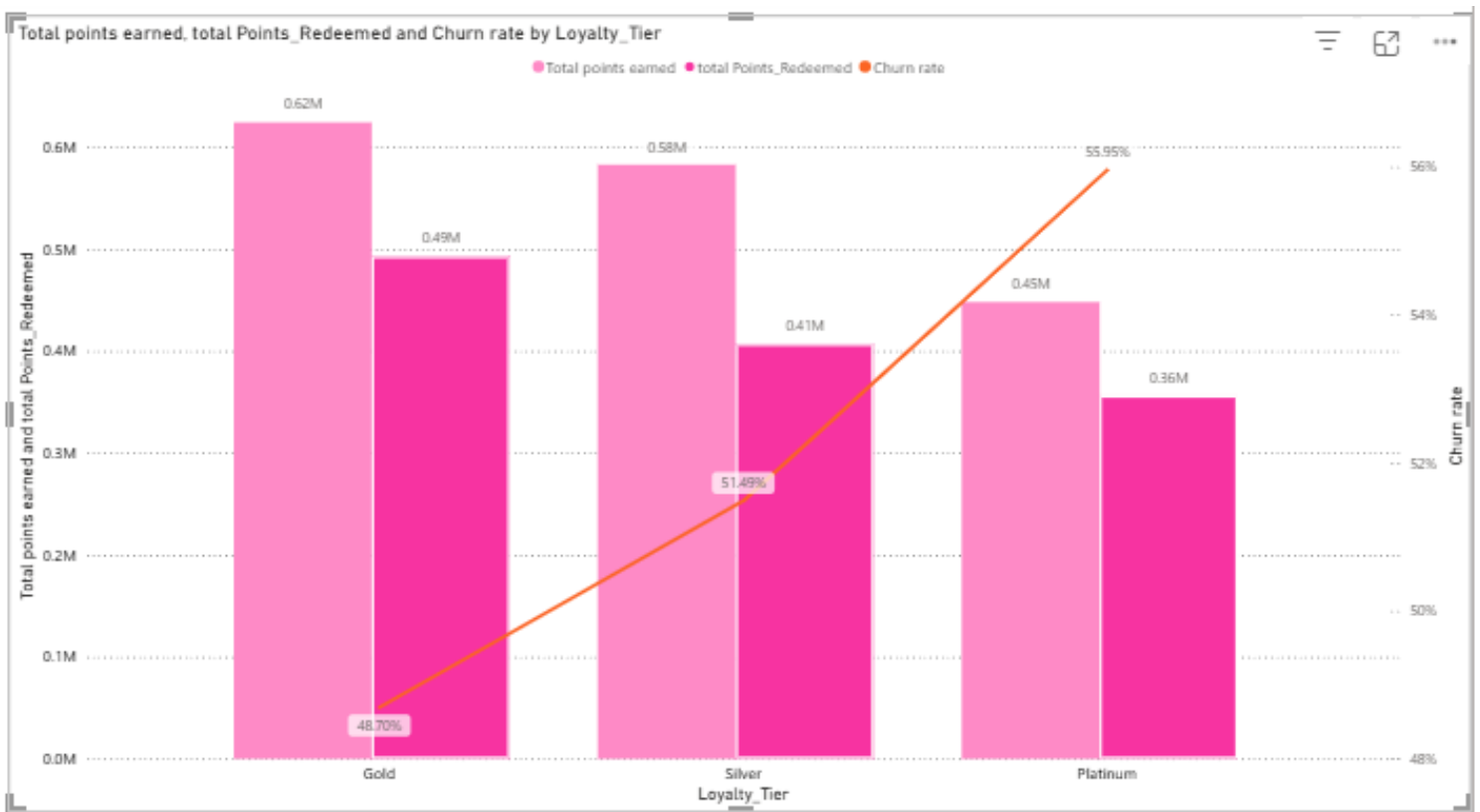
avg purchase amount (promo)

Loyalty_Tier	
Gold	Silver
Platinum	

2) Comparing churn rate Across loyalty tiers

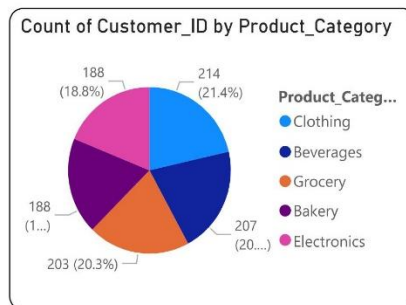
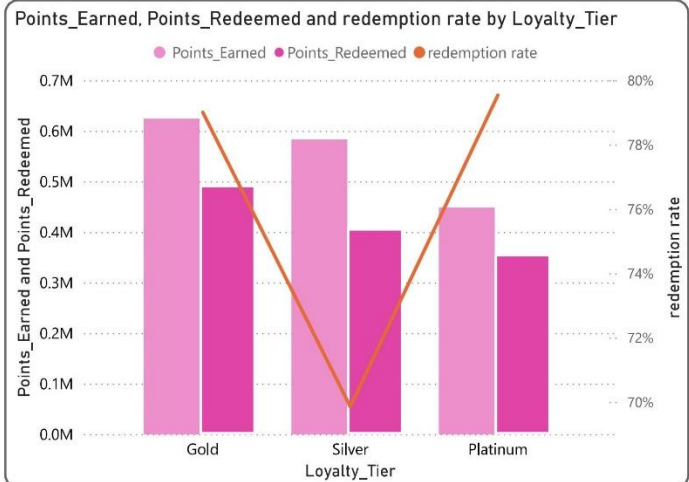


3) Points earned vs point redeemed By loyalty tiers



4) Recommendations For how to improve Redemption and retention

- * Churn rate was more than retention rate Only gold tier members Had more retention rate than churn rate
- * Customer having silver Loyalty Had the least amount of redemption rate for points
- * The most bought product category for Silver tier member was beverages So my suggestion would be To make sure that the beverages are available in each stores And has more promotion on it That may increase redemption rate And reduce churn rate For the silver tier But another observation showed that The average purchase amount During promotions was lower than The purchase amount without promotion for silver tier so some special discount And offers may help fix this issue
- * We also observed a general increase in average purchase amount During promotions That can also lead to overall Increase in Redemption rate of the points so tesco should focus on promos and deals



75.93%
redemption rate

51.67%
Churn rate

48.33%
retention rate

Task 5 (Store performance)

Objective: Store characteristics Two retention Outcome

Measures: Below are the measures used in this task

- ▼  Measure table
- ☐  average Purchase value
 - ☐  Churn rate
 - ☐  no of Churned labled Customer
 - ☐  non Churned customer
 - ☐  Repeat Customers
 - ☐  repeat rate
 - ☐  Retention Rate

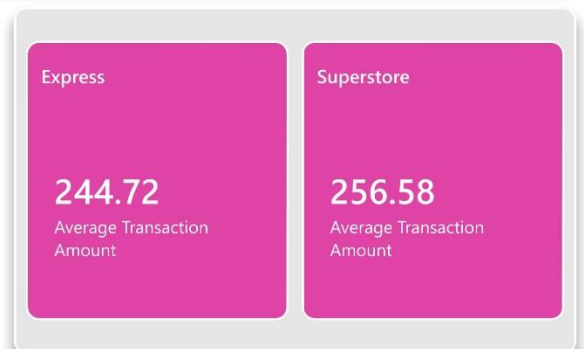
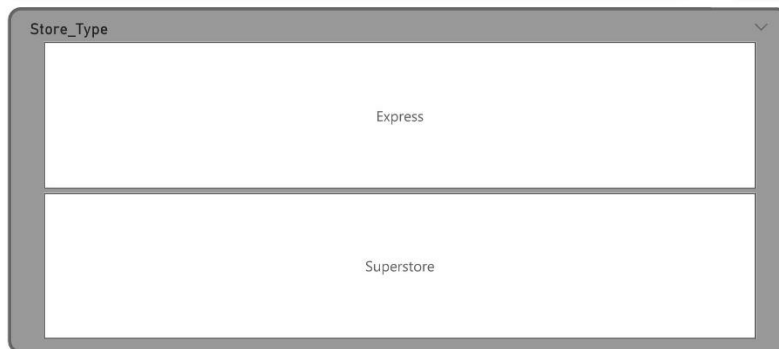
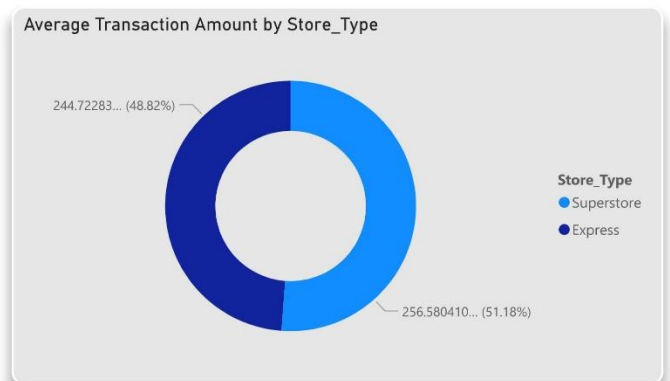
[Measures](#)

1) Margin columns From transaction table To store location table

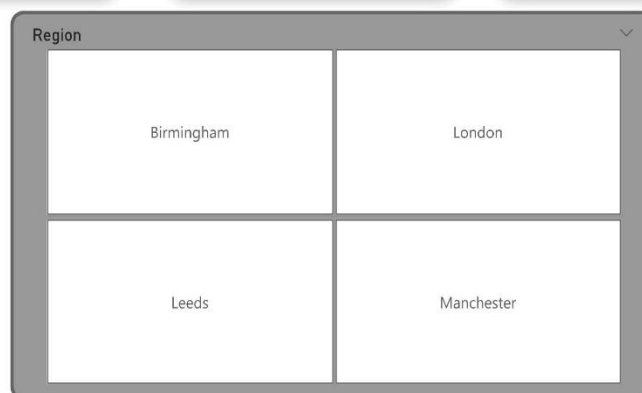
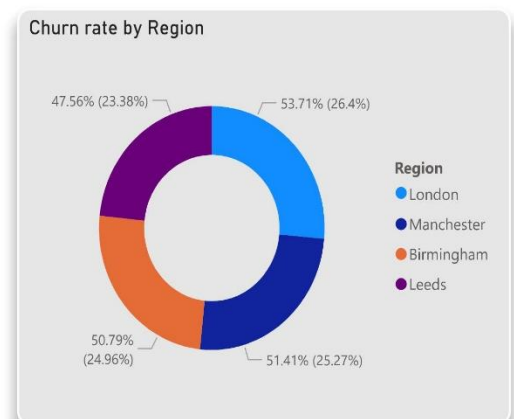
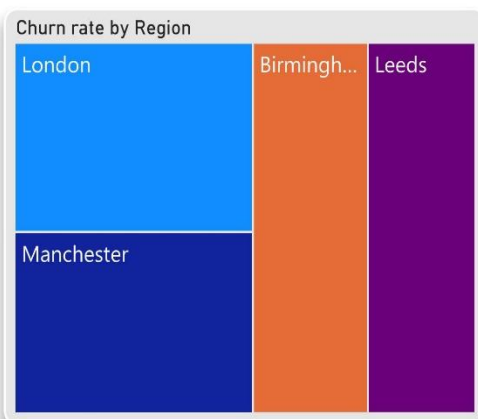
Manage relationships							
 New relationship		 Autodetect		 Edit		 Delete	
☐ From: table (column) ↑		Relationship		To: table (column)		Status	
☐	Customer_Demographics (1) (...)			Churn_Labelled_Customers (1)...		Active	...
☐	Customer_Transactions (1) (Cu...			Customer_Demographics (1) (...)		Active	...
☐	Customer_Transactions (1) (Sto...			Store_Locations (1) (Store_ID)		Active	...
☐	Loyalty_Program (1) (Custome...			Customer_Demographics (1) (...)		Active	...

2) Visualizing Stores by different metrics.

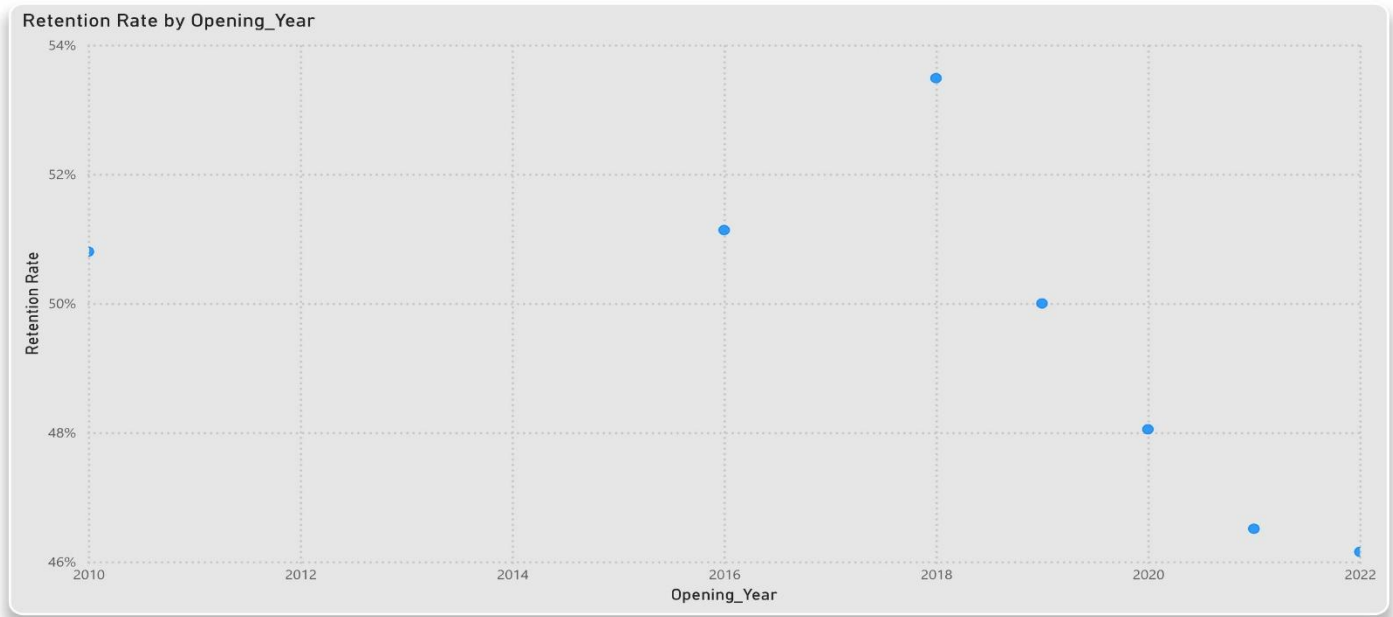
- Average transaction amount by store type



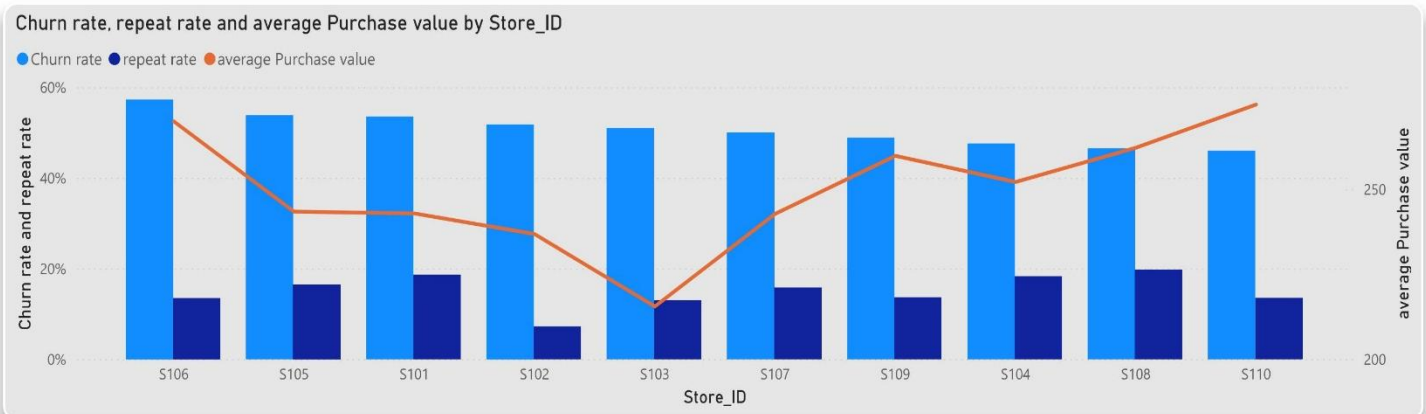
- Churn rate by store region



- Correlation between store opening year and retention



3) Suggestion for running store specific campaign



Churn rate, repeat rate and average Purchase value by Region

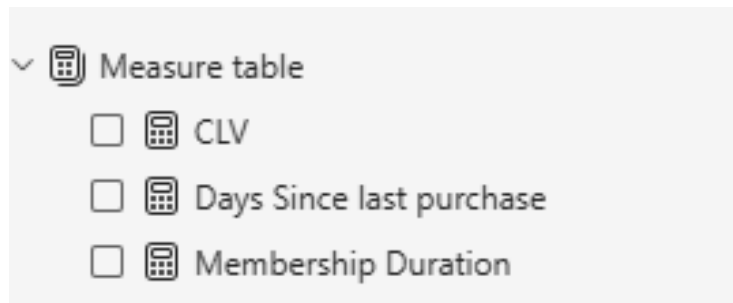


- * In terms of churn rate stores in London Had the highest number of Churn rate While also having the lowest amount of average purchase value
- * In terms of repeating customers Stores in Leeds have the lowest number of repeating customers conversion rate
- * If we look at the individual stores Store number S106 had the highest Churn rate
- * Store number S102 had the lowest Repeat rate
- * Store number S103 had the lowest amount Average purchase rate
- * In order to increase Repeating customers Tesco should Run campaigns at the stores in Manchester and Leeds
- * In order to Reduce the Churn rate Campaign should be Launched In London as priority
- * In order to increase average purchase value the camp Campaign should run In London followed by Leeds as priority

Task 6 (Identifying higher value customers)

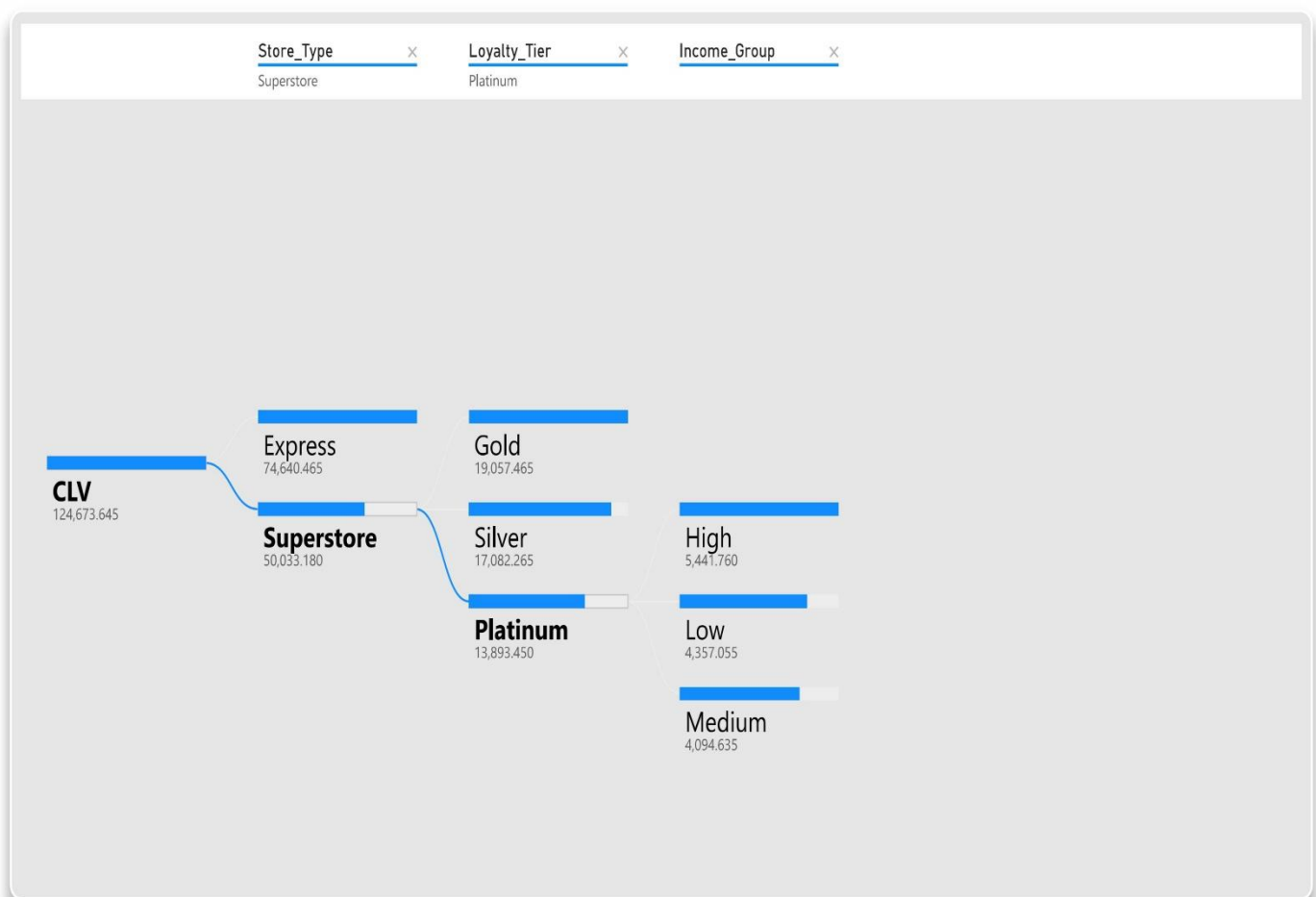
Objective: Identify and prioritize high value customer

Measures: Below are the measures used in this Task



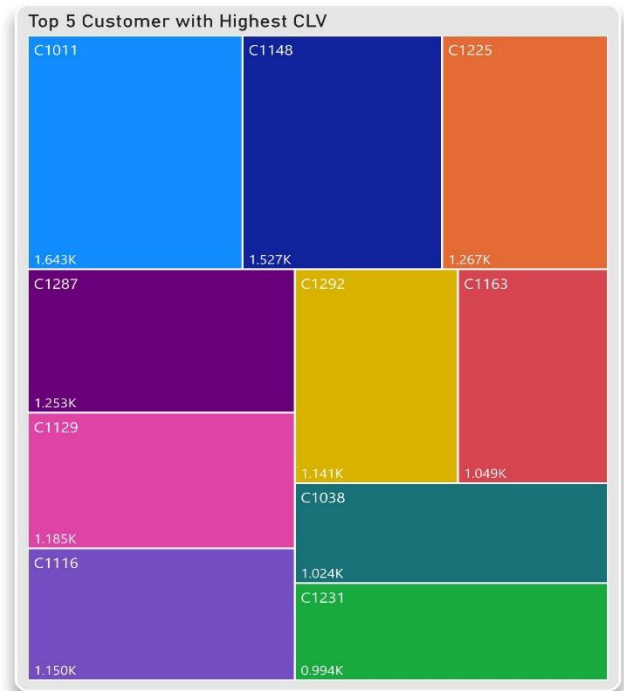
[Measures](#)

1) Calculating CLV and visualizing it .



2) Segmenting customer based on CLV.

Customer_ID	CLV	Sum of Amount	Membership Duration
C1287	1,252.830	2,505.66	2
C1038	1,024.485	2,048.97	2
C1078	980.650	1,961.30	2
C1263	940.695	1,881.39	2
C1265	886.875	1,773.75	2
C1225	844.973	2,534.92	3
C1148	763.603	3,054.41	4
C1163	699.593	2,098.78	3
C1132	669.980	1,339.96	2
C1231	662.600	1,987.80	3
C1011	657.072	3,285.36	5
C1187	650.873	1,952.62	3
C1009	648.135	1,296.27	2
C1036	627.515	1,255.03	2
C1152	609.933	1,829.80	3
C1116	575.035	2,300.14	4
C1292	570.313	2,281.25	4
C1207	565.560	1,131.12	2
C1007	564.307	1,692.92	3
C1172	548.315	1,096.63	2
C1270	531.745	1,063.49	2
C1072	528.480	1,585.44	3
C1141	520.095	1,040.19	2
C1081	510.260	1,530.78	3
C1182	483.235	966.47	2
C1129	474.088	2,370.44	5
C1025	472.655	945.31	2
C1262	468.737	1,406.21	3
C1160	466.333	1,399.00	3
Total	124,673.645	249,347.29	2



Background color - CLV

Format style

Rules

Apply to

Values only

What field should we base this on?

CLV

Rules

Reverse color order

New rule

If value

>=

0

Percent

and

<

25

Percent

then

If value

>=

25

Percent

and

<

75

Percent

then

If value

>=

75

Percent

and

<=

100

Percent

then

Learn more about conditional formatting

OK

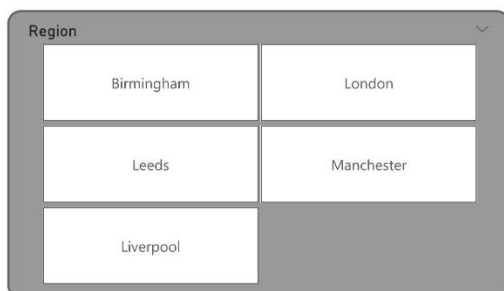
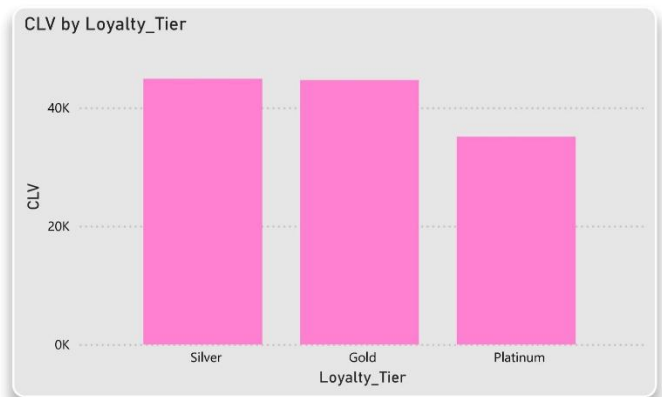
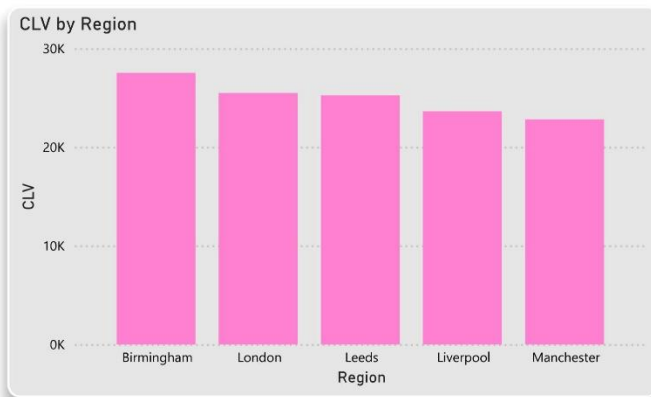
Cancel

3) Visualizing clv.

- CLV vs days since last purchase



- CLV by loyalty tier and region



Task 7 (Analysis and insight)

Objective: Communicate insights for business action

Measures: Below are the measures used in this table

- ▼  Measure Table
- ☐  % of Point earned vs redeemed
 - ☐  average Purchase amt.
 - ☐  average Purchase amt. w/o promo
 - ☐  average Purchase amt. with promo
 - ☐  avg. Quantity of item w Promo
 - ☐  avg. Quantity of item w/o Promo
 - ☐  churn_Rate
 - ☐  CLV
 - ☐  max Purchase amt.
 - ☐  Membership Duration
 - ☐  no of Churned labled Customer
 - ☐  Repeat churned Customers
 - ☐  repeat customer rate
 - ☐  Repeat Customer target
 - ☐  Repeat Customers
 - ☐  Sales Diff w/wo Promo
 - ☐  Sales With promo
 - ☐  Sales Without promo
 - ☐  Target Churn rate
 - ☐  Target Redemption Rate

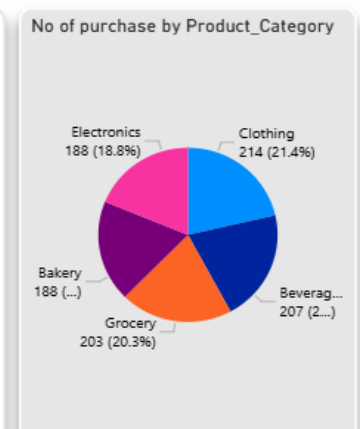
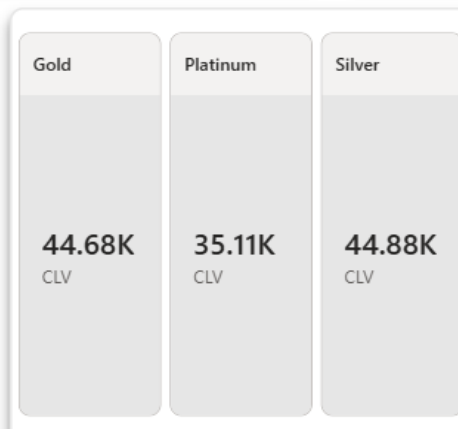
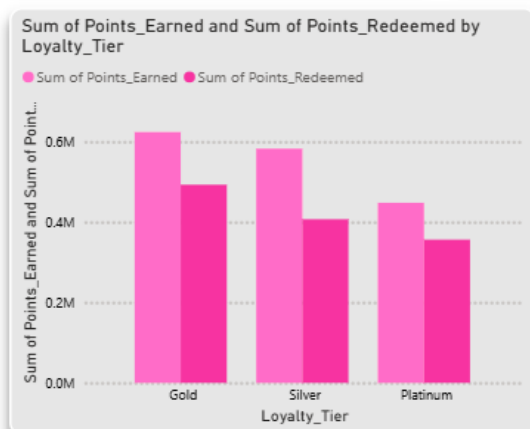
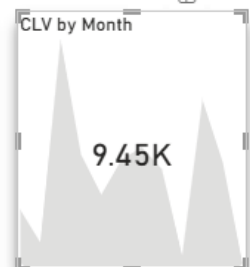
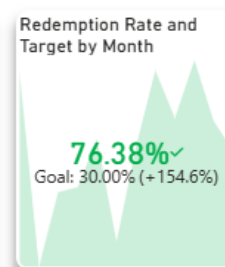
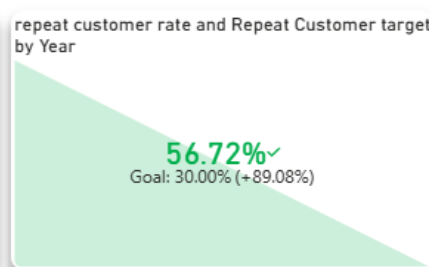
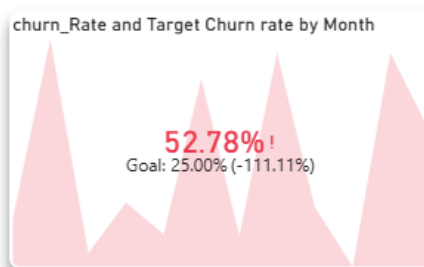
[Measures](#)

1) Creating multi page report

- Slicer



- Overview



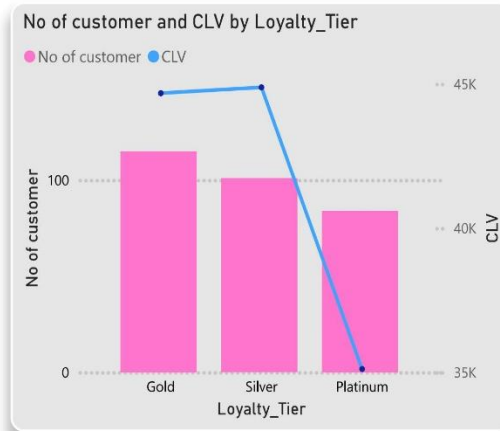
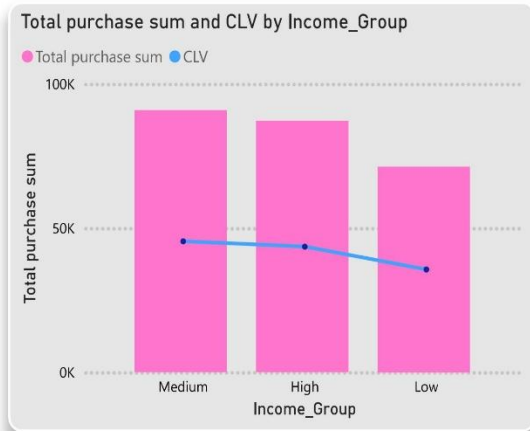
- Loyalty and promotion impact



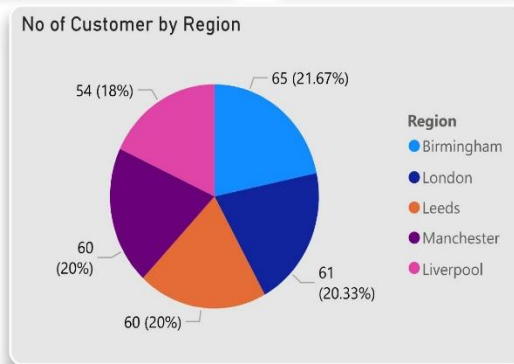
- Stores and region insight



- Customer segmentation



Customer	CLV
C1287	1,252.83
C1038	1,024.49
C1078	980.65
C1263	940.70
C1265	886.88
C1225	844.97
C1148	763.60
C1163	699.59
C1132	669.98
Total	124,673.64



2) Top three recommendations

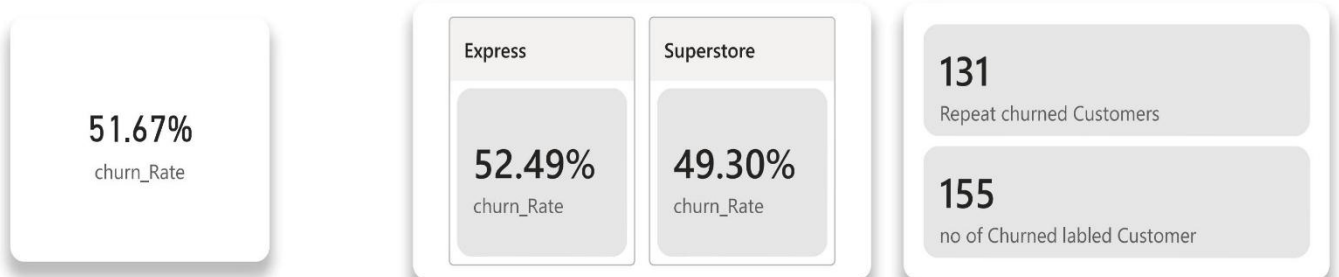
Top recommendations for Tesco from my sides are:

- That some stores have really high churn rate But the churn rate of store type superstores Is less than 50% That means We're retaining More than half of our customers And also the average purchase amount A superstores are higher than express type stores But the numbers of super stores Are significantly less than the number of expert stores So Tesco should focus Increasing the number of superstores With the focus of increasing Clv
- London stores create the highest generate on average Tesco should run Customer retention campaigns On priority basis In this region
- The repeat rate of express stores Are significantly higher Then of superstores that means The customers are repeatedly purchasing items But the churn rate in the express stores are more than 50% so that means that we are losing these repeated and valuable customers Tesco should focus on Running customer retention campaigns in express stores Or try to Increase the number of superstores
- We found out that the platinum tier remember At the highest churn rate and they were not affected by promotions And we also found out that during promotions The performance of purchases decreased So Tesco Should focus more on The platinum tier users

3) What should be done to retain more customer

Advice for customer retention In Tesco

- If the main goal is customer retention and reducing churn rate More super stores should be introduced with focus on increasing repeated customers as the repeat rate in superstores is less than express stores
- We saw that the promotion increase purchase value and quantity on average So Tesco should focus on promotion in higher churn segments
- Out of 155 Churned customer More than 130 customers were repeating customers If repeating customers are leaving Tesco Then Tesco needs to work on their Loyalty programme On priority basis And add new perks and features In their loyalty tier



4) Where should tesco focus next

What Tesco should focus on next:

- Tesco should focus on platinum tier and low income group customers as they are the least performing
- They should run campaigns for January reduction and retaining customers
- Super stores should be launched and enhanced for increasing repeat rate and further reducing Churn rate
- Tesco should also improve their promotion deals as it is effective On some Tire and income group but are not performing well on others
- Clothing and beverages are the top two best performing categories In majority of segments and should Be the main focus for promotions and deals

Measures

- no of Churned labled Customer = $\text{CALCULATE}(\text{COUNTROWS}('Churn_Labelled_Customers (1)'), 'Churn_Labelled_Customers (1)'[Churned (Yes/No)]="Yes")$
- Repeat Customers = $\text{CALCULATE}(\text{DISTINCTCOUNT}('Customer_Transactions (1)'[Customer_ID]), \text{FILTER}(\text{VALUES}('Customer_Transactions (1)'[Customer_ID]), \text{CALCULATE}(\text{COUNT}('Customer_Transactions (1)'[Customer_ID]))>1))$
- repeat customer rate = $\text{DIVIDE}([\text{Repeat Customers}], \text{DISTINCTCOUNT}('Customer_Transactions (1)'[Customer_ID]))$
- Repeat churned Customers = $\text{CALCULATE}([\text{Repeat Customers}], 'Churn_Labelled_Customers (1)'[Churned (Yes/No)]="Yes")$
- average Purchase amt. = $\text{AVERAGE}('Customer_Transactions (1)'[Amount])$
- Sales With promo = $\text{CALCULATE}(\text{SUM}('Customer_Transactions (1)'[Amount]), 'Customer_Transactions (1)'[Promotion_Applied]="Yes")$
- Sales Without promo = $\text{CALCULATE}(\text{SUM}('Customer_Transactions (1)'[Amount]), 'Customer_Transactions (1)'[Promotion_Applied]="No")$
- Sales Diff w/wo Promo = $\text{DIVIDE}([\text{Sales With promo}]-[\text{Sales Without Promo}], [\text{Sales Without Promo}])$
- Membership Duration = $\text{DATEDIFF}(\text{MAX}('Customer_Demographics (1)'[Membership_Since]), \text{TODAY}(), \text{YEAR})$
- CLV = $\text{DIVIDE}(\text{SUM}('Customer_Transactions (1)'[Amount]), [\text{Membership Duration}])$
- avg. Quantity of item w Promo = $\text{CALCULATE}(\text{AVERAGE}('Customer_Transactions (1)'[Quantity]), 'Customer_Transactions (1)'[Promotion_Applied]="Yes")$
- avg. Quantity of item w/o Promo = $\text{CALCULATE}(\text{AVERAGE}('Customer_Transactions (1)'[Quantity]), 'Customer_Transactions (1)'[Promotion_Applied]="No")$
- % of Point earned vs redeemed = $\text{DIVIDE}(\text{SUM}('Loyalty_Program (1)'[Points_Redeemed]), \text{SUM}('Loyalty_Program (1)'[Points_Earned]))$
- churn_Rate = $\text{DIVIDE}([\text{no of Churned labled Customer}], \text{COUNTROWS}('Churn_Labelled_Customers (1)'), 0)$
- max Purchase amt. = $\text{MAX}('Customer_Transactions (1)'[Amount])$
- average Purchase amt. with promo = $\text{CALCULATE}(\text{AVERAGE}('Customer_Transactions (1)'[Amount]), 'Customer_Transactions (1)'[Promotion_Applied]="Yes")$
- average Purchase amt. w/o promo = $\text{CALCULATE}(\text{AVERAGE}('Customer_Transactions (1)'[Amount]), 'Customer_Transactions (1)'[Promotion_Applied]="No")$
- Target Churn rate =0.25

- Repeat Customer target = 0.30
- Target Redemption Rate = 0.30
- no of customer in Churn table = COUNT('Churn_Labelled_Customers (1)'[Churned (Yes/No)])
- Repeat customer = COUNTROWS(FILTER(VALUES('Customer_Transactions (1)'[Customer_ID]),CALCULATE(COUNT('Customer_Transactions (1)'[Customer_ID])>1)))
- Low tear customer = CALCULATE(DISTINCTCOUNT('Customer_Transactions (1)'[Customer_ID]),FILTER(VALUES('Customer_Transactions (1)'[Customer_ID]),CALCULATE(COUNTROWS('Customer_Transactions (1)')) > 1 && CALCULATE(COUNTROWS('Customer_Transactions (1)')) < 5))
- mid tear customer = CALCULATE(DISTINCTCOUNT('Customer_Transactions (1)'[Customer_ID]),FILTER(VALUES('Customer_Transactions (1)'[Customer_ID]),CALCULATE(COUNTROWS('Customer_Transactions (1)')) > 4 && CALCULATE(COUNTROWS('Customer_Transactions (1)')) < 11))
- High tear customer = CALCULATE(DISTINCTCOUNT('Customer_Transactions (1)'[Customer_ID]),FILTER(VALUES('Customer_Transactions (1)'[Customer_ID]),CALCULATE(COUNTROWS('Customer_Transactions (1)')) > 10))
- Average Purchase frequency = DIVIDE(COUNTROWS('Customer_Transactions (1)'),DISTINCTCOUNT('Customer_Transactions (1)'[Customer_ID]))
- Repeat rate= DIVIDE([Repeat customer],DISTINCTCOUNT('Customer_Transactions (1)'[Customer_ID]))
- Average order quantity = AVERAGE('Customer_Transactions (1)'[Quantity])
- repeat customer during promo = CALCULATE([Repeat customer],'Customer_Transactions (1)'[Promotion_Applied]="Yes")
- Transaction with promotion = CALCULATE(COUNTROWS('Customer_Transactions (1)'),'Customer_Transactions (1)'[Promotion_Applied] = "Yes")
- Transaction with promotion % = DIVIDE([Transaction with promotion],COUNTROWS('Customer_Transactions (1)'))
- avg purchase amout (promo) = CALCULATE(AVERAGE('Customer_Transactions (1)'[Amount]),'Customer_Transactions (1)'[Promotion_Applied]="Yes")
- avg purchase amout (no promo) = CALCULATE(AVERAGE('Customer_Transactions (1)'[Amount]),'Customer_Transactions (1)'[Promotion_Applied]="No")
- Churned people = CALCULATE(COUNTROWS('Churn_Labelled_Customers (1)'),'Churn_Labelled_Customers (1)'[Churned (Yes/No)]="Yes")

- Non Churned people = $\text{CALCULATE}(\text{COUNTROWS}('Churn_Labelled_Customers (1)'), 'Churn_Labelled_Customers (1)'[Churned (Yes/No)] = "No")$
- Churn rate = $\text{DIVIDE}([\text{Churned people}], \text{COUNTROWS}('Customer_Transactions (1)'))$
- retention rate = $1 - [\text{Churn rate}]$
- redemption rate = $\text{DIVIDE}(\text{SUM}('Loyalty_Program (1)'[Points_Redeemed]), \text{SUM}('Loyalty_Program (1)'[Points_Earned]))$
- Retention Rate = $1 - [\text{Churn rate}]$
- average Purchase value = $\text{AVERAGE}('Customer_Transactions (1)'[Amount])$
- non Churned customer = $\text{CALCULATE}(\text{COUNT}('Churn_Labelled_Customers (1)'[Customer_ID]), 'Churn_Labelled_Customers (1)'[Churned (Yes/No)] = "No")$
- Days Since last purchase = $\text{MAX}('Churn_Labelled_Customers (1)'[Days_Since_Last_Purchase])$

Task 8 (video)

- <https://www.loom.com/share/857ee18c479a475a82365143c19a922d>
- I was not able to Download the video from the Loom and All the other software did not perform well on my system So this link is the online direct video link Thank you for understanding.
- It was Uncertain that I was not able to download The video I will make sure that I'm prepared with other software's in my photo projects